RAJ MEHTA

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EXPERIENCE

AngelList Jun 2024 - Present
Software Engineer - AI New York, NY

- Built an end-to-end RAG pipeline to extract and verify investment terms from complex financing documents using OCR, embedding-based chunk retrieval, and prompt-tuned LLMs - automating 75% of document intake.
- Designed a scalable document processing backend using AWS S3 for storage, DynamoDB for term lookups, OpenSearch for embedding retrieval, and SQS-based queuing to support high-throughput ingestion.
- Created dynamic table parsing modules to handle unstructured tabular layouts in PDFs, integrating custom heuristics with LLM parsing strategies to boost structured data extraction coverage.
- Developed custom LLM eval harnesses and observability tools, enabling traceable monitoring and continuous regression testing.
- Shipped multi-source internal agents that interface with Slack, CRM, and Guru APIs to surface context-aware summaries for customer success reps, decreasing email response latency by 20%.

Oscar Health May 2022 - Aug 2022
Software Engineer New York, NY

- Deployed a pipeline for generating and sending annual health coverage tax forms to policyholders as part of the Member Data API team; successfully developed custom PDF documents for over a million members and wrote automated auxiliary scripts.
- Created robust API endpoints to fetch member data and fill forms using Python, gRPC, and Protocol Buffers complete with end to end testing.
- Performed complex database queries using PostgreSQL and SQL Alchemy to extract necessary member data, improving retrieval efficiency.

Masimo Corporation May 2021 - Aug 2021
Machine Learning Engineer Remote

- Conceptualized and led development of a movement tracking device and a sleep tracking device for a medical device company.
- Leveraged MATLAB's signal processing toolbox and machine learning toolbox to analyze key gait phases and process EEG, heart rate, and accelerometer data to accurately determine sleep stages on a wearable device.

RESEARCH/PROJECTS

TheAgentCompany: LLM Agent Benchmark for Real-World Tasks (Link) Oct 2024 - Dec 2024

- Co-authored a paper submitted to NeurIps (under review) for evaluating LLM agents on consequential real-world workplace tasks within a simulated software company environment.
- Collaborated with Graham Neubig and the Language Technologies Institute at Carnegie Mellon University to design and implement 175 diverse professional tasks across various domains.

A Multimodal Model to Bridge Chemical Compounds with Natural Language Feb 2024 - May 2024

- Led a collaborative initiative at Weill Cornell to design and implement a multimodal cheminformatics model.
- Leveraged 'SELFIES' molecular representations to integrate molecular structures with textual data, enabling the generation and retrieval of pharmaceutical compounds.

EDUCATION

Cornell University (Tech Campus) Aug 2023 - May 2024
Master of Engineering in Computer Science | GPA: 3.85 New York, NY
Relevant Coursework: Deep Generative Models, Computer Systems and Architecture, Machine Learning Engineering

Carnegie Mellon University Aug 2019 - May 2023
Bachelor of Science in Cognitive Science with Honors, Minor in Machine Learning | GPA: 3.80 Pittsburgh, PA
Relevant Coursework: Data Structures and Algorithms, Functional Programming, Deep Learning, Reinforcement Learning, Multimodal Machine Learning, Linear Algebra, 3D-Calculus, Statistics, Discrete Mathematics

SKILLS

Programming Languages: Python, C, R, SQL, MATLAB, TypeScript, Standard ML, CSS, HTML
Technologies: PyTorch, Langchain, SciKit-learn, AWS, Git, Flutter, React, Pandas, Numpy,

LEADERSHIP

CMU Machine Learning Department, Head Teaching Assistant Aug 2022 - Apr 2023

- Led recitations to explain topics such as Monte Carlo Tree Search, Policy Gradients, and Actor-Critic methods for 100+ students.

JET Learn, Python Tutor Jan 2021 - Feb 2022

- Designed syllabus and instructed high school students to use Python libraries; enabled them to build games and applications.